

Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE): Determining Therapeutic Goals for Treat-to-Target

L. Peyrin-Biroulet, MD, PhD¹, W. Sandborn, MD², B.E. Sands, MD, MS³, W. Reinisch, MD^{4,5}, W. Bemelman, MD, PhD⁶, R.V. Bryant, MD⁷, G. D'Haens, MD, PhD⁸, I. Dotan, MD⁹, M. Dubinsky, MD¹⁰, B. Feagan, MD¹¹, G. Fiorino, MD, PhD¹², R. Gearry, MBChB, PhD¹³, S. Krishnareddy, MD¹⁴, P.L. Lakatos, MD¹⁵, E.V. Loftus Jr, MD¹⁶, P. Marteau, MD, PhD¹⁷, P. Munkholm, MD¹⁸, T.B. Murdoch, MD, MSc, FRCPC¹⁹, I. Ordás, MD, PhD²⁰, R. Panaccione, MD, FRCPC²¹, R.H. Riddell, MD²², J. Ruel, MD, FRCPC²³, D.T. Rubin, MD²⁴, M. Samaan, MBBS⁸, C.A. Siegel, MD²⁵, M.S. Silverberg, MD, PhD²⁶, J. Stoker, MD, PhD²⁷, S. Schreiber, MD, PhD²⁸, S. Travis, DPhil, FRCPC²⁹, G. Van Assche, MD, PhD^{30,31}, S. Danese, MD, PhD¹², J. Panes, MD, PhD²⁰, G. Bouguen, MD³², S. O'Donnell, MD²⁶, B. Pariente, MD, PhD³³, S. Winer, MD, PhD³⁴, S. Hanauer, MD³⁵ and J.-F. Colombel, MD³

- OBJECTIVES:** The Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE) program was initiated by the International Organization for the Study of Inflammatory Bowel Diseases (IOIBD). It examined potential treatment targets for inflammatory bowel disease (IBD) to be used for a “treat-to-target” clinical management strategy using an evidence-based expert consensus process.
- METHODS:** A Steering Committee of 28 IBD specialists developed recommendations based on a systematic literature review and expert opinion. Consensus was gained if $\geq 75\%$ of participants scored the recommendation as 7–10 on a 10-point rating scale (where 10=agree completely).
- RESULTS:** The group agreed upon 12 recommendations for ulcerative colitis (UC) and Crohn's disease (CD). The agreed target for UC was clinical/patient-reported outcome (PRO) remission (defined as resolution of rectal bleeding and diarrhea/altered bowel habit) and endoscopic remission (defined as a Mayo endoscopic subscore of 0–1). Histological remission was considered as an adjunctive goal. Clinical/PRO remission was also agreed upon as a target for CD and defined as resolution of abdominal pain and diarrhea/altered bowel habit; and endoscopic remission, defined as resolution of ulceration at ileocolonoscopy, or resolution of findings of inflammation on cross-sectional imaging in patients who

¹INSERM Unité 954 and Department of Hepato-Gastroenterology, University Hospital of Nancy, University of Lorraine, Houdemont, France; ²Division of Gastroenterology, University of California San Diego, La Jolla, California, USA; ³Henry D. Janowitz Division of Gastroenterology, Icahn School of Medicine at Mount Sinai, New York, New York, USA; ⁴McMaster University, Hamilton, Ontario, Canada; ⁵Medical University of Vienna, Vienna, Austria; ⁶Department of Surgery, Academic Medical Center, Amsterdam, The Netherlands; ⁷Translational Gastroenterology Unit, John Radcliffe Hospital, Oxford, UK; ⁸Department of Gastroenterology, Academic Medical Centre, Amsterdam, The Netherlands; ⁹Inflammatory Bowel Disease Center, Department of Gastroenterology and Liver Diseases, Tel Aviv Sourasky Medical Center, Sackler Faculty of Medicine, Tel Aviv University, Tel Aviv, Israel; ¹⁰Division of Pediatric Gastroenterology and Hepatology, Icahn School of Medicine at Mount Sinai, New York, New York, USA; ¹¹University of Western Ontario, London, Ontario, Canada; ¹²Inflammatory Bowel Disease Center, Department of Gastroenterology, Humanitas Research Hospital, Rozzano, Milan, Italy; ¹³Department of Medicine, University of Otago, Christchurch, New Zealand; ¹⁴Columbia University Division of Digestive and Liver Diseases, New York, New York, USA; ¹⁵First Department of Medicine, Semmelweis University, Budapest, Hungary; ¹⁶Division of Gastroenterology and Hepatology, Mayo Clinic, Rochester, Minnesota, USA; ¹⁷AP-HP, Department of Digestive Diseases, Hôpital Lariboisière Medico-surgical and University Denis Diderot, Paris, France; ¹⁸North Zealand Hospital, Capital Region, University of Copenhagen, Copenhagen, Denmark; ¹⁹Inflammatory Bowel Disease Group, Mount Sinai Hospital, University of Toronto, Toronto, Ontario, Canada; ²⁰Department of Gastroenterology, Hospital Clinic of Barcelona, IDIBAPS, Centro de Investigación Biomédica en Red de Enfermedades Hepáticas y Digestivas (CIBEREHD), Barcelona, Spain; ²¹Inflammatory Bowel Disease Clinic University of Calgary, Calgary, Alberta, Canada; ²²Department of Pathology and Laboratory Medicine, Mount Sinai Hospital, Toronto, Ontario, Canada; ²³Division of Gastroenterology, CHU Sherbrooke, Sherbrooke, Quebec, Canada; ²⁴Section of Gastroenterology, Hepatology and Nutrition, University of Chicago Medicine, Chicago, Illinois, USA; ²⁵Dartmouth-Hitchcock Inflammatory Bowel Disease Center, Hanover, New Hampshire, USA; ²⁶Zane Cohen Centre for Digestive Diseases, Division of Gastroenterology, Mount Sinai Hospital, University of Toronto, Toronto, Ontario, Canada; ²⁷Department of Radiology, Academic Medical Center, University of Amsterdam, Amsterdam, The Netherlands; ²⁸Department of General Internal Medicine, Christian-Albrechts-University, University Hospital Schleswig Holstein, Kiel, Germany; ²⁹Translational Gastroenterology Unit, John Radcliffe Hospital, Oxford, UK; ³⁰Division of Gastroenterology, University of Leuven, Leuven, Belgium; ³¹University of Toronto, Toronto, Ontario, Canada; ³²Service des Maladies de l'Appareil Digestif et INSERM U991, CHU Pontchaillou et Université de Rennes 1, Rennes, France; ³³Department of Hepato-Gastroenterology, Claude Huriez Hospital, Université Lille Nord de France, Lille, France; ³⁴Department of Laboratory Medicine and Pathobiology, University of Toronto, Toronto, Ontario, Canada; ³⁵Northwestern University Feinberg School of Medicine, Chicago, Illinois, USA. **Correspondence:** S. Hanauer, MD, Northwestern University Feinberg School of Medicine, 676N St Clair—Suite 1400, Chicago, Illinois 60611, USA or J.-F. Colombel, MD, Henry D. Janowitz Division of Gastroenterology, Icahn School of Medicine at Mount Sinai, 1468 Madison Avenue, New York, New York 10029, USA. E-mail: shanauer@northwestern.edu or jean-frederic.colombel@mssm.edu

Received 1 December 2014; accepted 13 June 2015

cannot be adequately assessed with ileocolonoscopy. Biomarker remission (normal C-reactive protein (CRP) and calprotectin) was considered as an adjunctive target.

CONCLUSIONS: Evidence- and consensus-based recommendations for selecting the goals for treat-to-target strategies in patients with IBD are made available. Prospective studies are needed to determine how these targets will change disease course and patients' quality of life.

SUPPLEMENTARY MATERIAL is linked to the online version of the paper at <http://www.nature.com/ajg>

Am J Gastroenterol 2015; 110:1324–1338; doi:10.1038/ajg.2015.233; published online 25 August 2015

INTRODUCTION

Inflammatory bowel diseases (IBD) are chronic, progressive, and disabling conditions. Most current strategies, which target control of symptoms, do not appear to significantly alter the natural course of the disease (1–5), although reductions in the need for surgery or the occurrence of neoplasia have been reported in some ulcerative colitis (UC) and Crohn's disease (CD) population-based cohorts over time (6–8). Recent studies underscore the need to look beyond symptoms and to treat endoscopic/macrosopic lesions in both CD and UC with the final aim of preventing structural damage and disability (9–12). Whether histologic healing should be an ultimate therapeutic goal is debated (11,13).

It is not feasible to repeat frequent endoscopies or cross-sectional imaging for reasons of cost and/or the invasive nature of these procedures; therefore, surrogate biomarkers of inflammation, including C-reactive protein (CRP) and fecal calprotectin have been increasingly studied in IBD, although their ability to replace endoscopic, histologic, and cross-sectional imaging studies remains ambiguous (14–16).

Acknowledging the need to accurately measure both the patient's experience with the disease and biological manifestations of disease, the US Food and Drug Administration (FDA) is moving from the Crohn's Disease Activity Index (CDAI) as a clinical trial end point toward patient-reported outcomes (PROs) and objective measures of disease, such as findings from endoscopy (17). The European Medicines Agency has not adopted such recommendations as of yet.

A "treat-to-target" strategy has been advocated as an optimized management approach for rheumatoid arthritis (RA). This approach is focused on achieving remission or low disease activity, based on insights from various clinical trials as identified through a systematic literature review (18). The purpose of strict definitions of treatment targets is to facilitate strategic actions in routine clinical practice. Accordingly, physicians and patients should discuss such targets and work toward achieving them by adopting therapeutic changes within distinct time frames, ideally following therapeutic algorithms (19). This approach has been used in many areas of medicine, where treatment targets have been defined to improve outcomes and reduce the risk of end-organ damage (such as joint damage in RA and macrovascular or microvascular complications of diabetes mellitus) (20–25).

The objective of the Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE) initiative, under the auspices of the International Organization for the Study of Inflammatory

Bowel Diseases (IOIBD), was to achieve international expert consensus on appropriate evidence-based treatment targets for IBD that could be used in treat-to-target strategies in routine clinical practice.

METHODS

This initiative by the IOIBD comprises several steps. First, a Steering Committee (SC) selected an Operational Committee consisting of 33 specialists (primarily IOIBD members who were evenly distributed across the globe) with expertise in treating IBD. Members of the Operational Committee were selected based on their training and experience; participation in clinical trials and clinical outcome research; and previous participation in the development of consensus statements.

Systematic literature review and generation of draft recommendations

The SC defined a series of broad search questions (What is the target? How can it be measured? Is the target already used as a target in clinical studies/trials? What are the data supporting the proposed target?) and selected six potential groups of treatment targets (clinical, endoscopic, histological, imaging, biomarkers, and PROs) to enable a comprehensive systematic literature review, full details of which will be published separately. Based on these questions, a search string was developed, tested, and refined for each of the six groups of treatment targets. PubMed, EMBASE, the Cochrane Library, and congress abstracts were searched. No search limits were set for full publications, other than specifying articles published in English. For congress abstracts, a date limit was set to retrieve abstracts published from 2008 to 2013. Details of the search terms and search results are given in **Supplementary Table 1** online.

The focus of the literature review was divided among six groups of gastroenterology fellows, each group mentored by two lead members from the SC and assigned responsibility for one of the six groups of potential treatment targets.

The fellows produced a summary report of the evidence, answering the aligned question. A level of evidence was assigned to each statement, in accordance with the Oxford Centre for Evidence-Based Medicine (CEBM) 2009 criteria (<http://www.cebm.net/oxford-centre-evidence-based-medicine-levels-evidence-march-2009/>) (26). Based on the results of their literature search, the fellows proposed a set of draft recommendations that were reviewed and modified by the lead SC members.

Voting and consensus meeting

On 3 April 2014, a consensus meeting preceding the annual IOIBD meeting in Amsterdam was convened that comprised 28 attendees, including members of the SC, the work-group leaders and eight fellows. The purpose of this meeting was to generate proposed statements for use in treat-to-target strategies in daily practice. This was achieved through discussion in breakout and plenary sessions. Each statement was then voted upon in an anonymous manner using a digital system.

Composite target statements, consisting of a combination of specific proposed targets arising from each workstream focus (e.g., endoscopy, imaging, and biomarkers), were developed and voted on, along with requisite methodologies and measurement tools.

Agreement was achieved if at least 75% of participants scored the statement within the 7–10 range on a 10-point numerical rating scale (1=do not agree at all, 10 agree completely). If a 75% consensus was not achieved, further discussion and amendment of voting statements was required, followed by a second round of voting. If no agreement was reached after two rounds of voting, then the statement was “excluded”.

RESULTS

In total, 15,681 references were retrieved during the literature search; 905 of these were analyzed and used to develop the draft recommendations (**Supplementary Table 1**). Statements for which there was agreement for CD and UC are summarized in **Tables 1** and **2**, respectively, together with an overview of key discussion points that arose in relation to the voting statements during the consensus meeting. The statements are not listed in a hierarchical order but rather within treatment target “categories”. The weight of individual statements is reflected by the strength of the recommendation/level of agreement as presented in the tables. The composite statements for CD and UC are shown in **Tables 1** and **2**, respectively (mean score for strength of recommendation: 7.7 (UC) and 8.0 (CD)). There was unanimous agreement that the underlying principle for therapy of IBD, whether CD or UC, should be control of inflammation. Statements that required amendment before agreement or that were excluded (consensus of $\geq 75\%$ not achieved) are shown in **Supplementary Table 2**.

Crohn's disease

Clinical targets. The clinical target is resolution of abdominal pain and normalization of bowel habit. However, resolution of symptoms alone is not a sufficient target, and objective assessment of inflammation of the bowel is necessary when making clinical decisions. Mucosal healing was recommended as the therapeutic goal in clinical practice, since it is associated with better outcomes in both cohort studies and randomized controlled trials. Therefore, the term *remission* should include both resolution of symptoms and mucosal inflammation.

Remission can be assessed using the CDAI, which has been developed using a prospective multistep approach and has been widely used in many clinical trials. Clinical remission is defined as a score below 150, and clinical response is defined as a decrease of

at least 70 points (although improved discrimination vs. placebo is observed with a decrease of 100 points) (27,28). The Harvey Bradshaw Index used in some clinical trials has not been prospectively validated, but is strongly correlated with the CDAI (29). The definition of clinical remission and response as assessed by Harvey Bradshaw Index is a score of ≤ 4 points and a 3-point decrease, respectively (30). In the pediatric population, the Pediatric CDAI is a validated score used in clinical trials (31,32) (CEBM 1).

Overall, clinical scoring systems remain poorly correlated with objective markers of inflammation, such as endoscopic assessment, or CRP to assess disease activity and response to treatment. In the Study of Biologic and Immunomodulator Naive Patients In Crohn's Disease trial, half the patients under azathioprine and/or infliximab in clinical remission had endoscopic and/or CRP evidence of residual active CD, whereas other patients with endoscopic and CRP normalization had persistent clinical symptoms (33). Thus, none of the existing clinical scoring systems is sufficiently accurate to predict mucosal inflammation as measured by CRP or endoscopy during CD (33–35) (CEBM 1). However, while assessment of clinical parameters alone may lack the necessary precision for decision making, it is still necessary to monitor and treat symptoms (36). The development of independent clinical measures that reliably assess symptoms from the patient's perspective is necessary and is discussed further in the section on PROs.

In clinical trials, steroid-free clinical remission is achievable and sustainable (37–39) (CEBM 1).

Endoscopic targets. The group agreed that the absence of ulceration is the most appropriate endoscopic target, and that endoscopic or, where endoscopy cannot be performed, cross-sectional imaging assessment should be performed 6–9 months after the start of therapy. Standard measures of endoscopic disease activity may include the CD Endoscopic Index of Severity (CDEIS) and the Simple Endoscopic Score for CD (SES-CD), with the SES-CD providing a practical and simple alternative to the CDEIS. The values of the two measurements correlate well with one another, and the inter- and intra-observer variability has been shown to be less for SES-CD scores than for CDEIS in a study of expert central readers (40,41) (CEBM 2b). It remains to be investigated, however, which degree of mucosal improvement is associated with clinical benefit.

For the assessment of remission using the CDEIS, a score of <3 indicates endoscopic remission, a score of ≥ 3 indicates active disease, and a decrease in CDEIS by >5 demonstrates response (16,42–44) (CEBM 2a). Using relative (rather than absolute) changes in SES-CD and CDEIS scores, such as a decrease from baseline of at least 50%, has been shown to have predictive value for clinical outcomes (45) (CEBM 2b). However, while such scores are available, they are not systematically used in clinical practice.

The Rutgeerts score may be used to assess postoperative recurrence, with a score of i2 and above indicating recurrence (46). However, this score has not been validated, and further investigation into the relevance of the i2 grade is needed, including

Table 1. Twelve recommendations for treating to target in Crohn's disease (CD) by the International Organization for the Study of Inflammatory Bowel Diseases (IOIBD)

Recommendation	Voting results	
	Strength of recommendation ^a	Percentage votes at last ballot
<i>Clinical</i>		
1. Resolution of abdominal pain and normalization of bowel habit should be the target	7.9	88
2. Resolution of symptoms alone is not a sufficient target. Objective evidence of inflammation of the bowel is necessary when making clinical decisions	8.2	91
All attendees agreed that resolution of clinical symptoms and inflammation are the goals of treatment that define the term "remission". Mucosal healing was recommended as the therapeutic goal in clinical practice, because it is associated with better outcomes in both cohort studies and randomized controlled trials.		
<i>Endoscopic</i>		
1. Absence of ulceration is the target	7.9	88
2. Resolution of symptoms alone is not a sufficient target. Objective evidence of inflammation of the bowel is necessary when making clinical decisions	8.2	91
<i>Histologic</i>		
1. Histologic remission is not a target	8.7	100
The attendees felt that the level of evidence was insufficient to recommend histologic remission as a target in CD in clinical practice		
<i>Imaging</i>		
1. When endoscopy cannot adequately evaluate inflammation, resolution of inflammation as assessed by cross-sectional imaging is a target	7.9	92
Endoscopy is a more mature instrument for defining mucosal healing, and experience of defining normalization of inflammation using imaging is less well established. The attendees agreed that cross-sectional imaging is not a universal target, although they acknowledged its role in patients with CD who are not adequately assessed by colonoscopy.		
<i>Biomarkers</i>		
1. Available biomarkers including CRP and fecal calprotectin are not targets	8.2	96
2. CRP and fecal calprotectin are adjunctive measures of inflammation for monitoring in CD. Failure of CRP or fecal calprotectin normalization (below lab-specific cutoff) should prompt further endoscopic evaluation, irrespective of symptoms.	8.4	100
The attendees concluded that biomarkers are not a target, because there is insufficient evidence to recommend treatment optimization by taking into account biomarkers alone. Biomarkers may reflect residual intestinal inflammation, but endoscopy and/or magnetic resonance imaging should be performed before optimizing medical treatment. As such, biomarkers facilitate the monitoring of a patient rather than being a target for treatment <i>per se</i> .		
<i>Patient-reported outcomes</i>		
1. The primary PRO for CD should be resolution of abdominal pain and normalization of bowel habit	8.1	83
2. In addition to resolution of abdominal pain and altered bowel habit, the patient's individual goals should also be addressed	8.2	96
3. The frequency of outcome assessment should be tailored to the patient's symptoms, with a minimum frequency of every 3 months until resolution. The frequency of outcome assessment after symptom resolution should be at least every 6–12 months	8.2	100
PROs are extensive and varied. The ultimate PRO is the normalization of quality of life (compared with before the onset of disease), achieved through the resolution of symptoms and inflammation. Acknowledging the great variability in the expression of symptoms, particularly in CD but also in UC, other individual symptoms that are affecting the patient's quality of life must be addressed; the goals of treatment should be tailored and agreed with the individual patient.		
<i>Composite end point</i>		
1. Clinical/PRO remission defined as resolution of abdominal pain and diarrhea/altered bowel habit, which should be assessed at a minimum of 3 months during active disease, and endoscopic remission defined as resolution of ulceration at ileocolonoscopy (or resolution of findings of inflammation on cross-sectional imaging in patients who cannot be adequately assessed with ileocolonoscopy), which should be assessed at 6- to 9-month intervals during the active phase	8.0	92
CRP, C-reactive protein; PRO, patient-reported outcome; UC, ulcerative colitis.		
^a Level of agreement on a scale of 1–10.		

Table 2. Twelve recommendations for treating to target in ulcerative colitis (UC) by the International Organization for the Study of Inflammatory Bowel Diseases (IOIBD)

Recommendation	Voting results	
	Strength of recommendation ^a	Percentage votes at last ballot
<i>Clinical</i>		
1. Resolution of rectal bleeding and normalization of bowel habit should be the target	8.1	95
2. Resolution of symptoms alone is not a sufficient target. Objective evidence of inflammation of the bowel is necessary when making clinical decisions	7.4	75
All attendees agreed that resolution of clinical symptoms and inflammation are the goals of treatment that define the term “remission”. Mucosal healing was recommended as the therapeutic goal in clinical practice, because it is associated with better outcomes in both cohort studies and randomized controlled trials.		
<i>Endoscopic</i>		
1. A Mayo endoscopic subscore of 0 is the optimal target. A Mayo endoscopic subscore of 1 should be a minimum target	7.8	92
2. Endoscopic assessment should be performed 3–6 months after the start of therapy for a patient with symptoms	8.0	92
The endoscopic Mayo subscore is currently recommended for UC, rather than the Ulcerative Colitis Endoscopic Index of Severity (UCEIS), because its predictive value is well established. While a Mayo subscore of 0 may be defined as the target, there is currently insufficient evidence to recommend it in all patients and only a Mayo subscore of 0–1 can be systematically recommended in clinical practice.		
<i>Histologic</i>		
1. Histopathology is a sensitive measure of inflammation but is not a target due to lack of evidence of clinical utility	7.5	80
The attendees felt that the level of evidence was insufficient to recommend histologic remission as a target in UC in clinical practice. However, it was considered important to acknowledge the role of histopathology in the evaluation of inflammation in patients with UC.		
<i>Imaging</i>		
1. Cross-sectional imaging is not a target in UC	8.4	92
<i>Biomarkers</i>		
1. Available biomarkers including CRP and fecal calprotectin are not targets	8.3	90
2. CRP and fecal calprotectin are adjunctive measures of inflammation for monitoring in UC. Failure of CRP or fecal calprotectin normalization (below lab-specific cutoff) should prompt further endoscopic evaluation, irrespective of symptoms	8.2	95
The attendees concluded that biomarkers are not a target, because there is insufficient evidence to recommend treatment optimization by taking into account biomarkers alone. Biomarkers may reflect residual intestinal inflammation, but endoscopy and/or magnetic resonance imaging should be performed before optimizing medical treatment. As such, biomarkers facilitate the monitoring of a patient rather than being a target for treatment <i>per se</i> .		
<i>Patient-reported outcomes</i>		
1. The primary PRO for UC should be resolution of rectal bleeding and normalization of bowel habit	8.1	83
2. In addition to resolution of rectal bleeding and altered bowel habit, the patient's individual goals should also be addressed	8.2	96
3. The frequency of outcome assessment should be tailored to the patient's symptoms, with a minimum frequency of every 3 months until resolution. The frequency of outcome assessment after symptom resolution should be at least every 6–12 months	8.2	100
PROs are extensive and varied. The ultimate PRO is the normalization of quality of life (compared with before the onset of disease), achieved through the resolution of symptoms and inflammation. Acknowledging the great variability in the expression of symptoms, particularly in CD but also in UC, other individual symptoms that are affecting the patient's quality of life must be addressed; the goals of treatment should be tailored and agreed with the individual patient.		
<i>Composite end point</i>		
1. Clinical/PRO remission defined as resolution of rectal bleeding and diarrhea/altered bowel habit, which should be assessed at a minimum of 3 months during active disease, and endoscopic remission defined as resolution of friability and ulceration at flexible sigmoidoscopy or colonoscopy, which should be assessed at 3-month intervals during the active phase	7.7	92
CD, Crohn's disease; CRP, C-reactive protein; PRO, patient-reported outcome.		
^a Level of agreement on a scale of 1–10.		

modification of its definition, to include an i2a (purely anastomotic lesions) and an i2b (lesions in the neoterminal ileum >5 aphthous ulcers) grade, where only i2b and greater would be considered as relevant mucosal recurrence (47,48) (CEBM 5).

Histological targets. The premise for histological healing in CD is that persistent inflammation confers an increased risk of persistent symptoms or clinical relapse, hospitalization, resection, and neoplasia, all of which lead to cumulative disability (49–56) (CEBM 4). Several definitions of histological remission in CD have been proposed, although the most appropriate goal is yet to be determined (57–59) (CEBM 5). The risk of sampling error creates additional challenges and histological assessments should take this into consideration. Since there are relatively few data for histological healing as a therapeutic end point, the group agreed that histological remission is not a treatment target for CD in clinical practice.

A paradigm shift in clinical thinking is required to consider incorporating histopathology into routine patient management in the future (CEBM 5). Currently, the Geboes system is used in clinical studies and clinical practice; however, better validation of these criteria is required (60) (CEBM 5). In addition, other clinical and molecular methods of assessing histological remission ideally should be developed and titrated against histopathology for long-term patient assessment (CEBM 5).

Imaging targets. Imaging is less well established than endoscopy for defining normalization of inflammation, and attendees agreed that resolution of lesions in cross-sectional imaging examinations is not a universal target, although imaging techniques may have a complementary role in patients with CD who are not adequately assessed by colonoscopy. The overall diagnostic accuracy of all cross-sectional imaging modalities (ultrasonography, magnetic resonance imaging and computed tomography enterography) for the assessment of disease activity and severity in CD is similar (higher than 80%) (61–65) (CEBM 1). The most established instrument for magnetic resonance imaging is the Magnetic Resonance Index of Activity, which scores wall thickness, relative contrast enhancement, mural edema and ulcers in different segments of the gastrointestinal tract (66–68). Although magnetic resonance enterography and computed tomography enterography have been shown to impact on therapeutic management in patients with CD (69–71) (CEBM 4), endoscopy may be required as a complementary tool to cross-sectional imaging techniques for the assessment of proximal lesions in the small bowel (72) (CEBM 2).

Biomarkers. Attendees agreed that available biomarkers, including CRP and fecal calprotectin, are not targets because there is insufficient evidence to recommend treatment optimization using biomarkers alone.

Overall, the best evidence is available for CRP. The literature shows that normalization of CRP is associated with therapeutic response in CD, and CRP levels correlate modestly with endoscopic disease activity (34,73–78) (CEBM 1b). Furthermore,

elevation in CRP at baseline predicts response to anti-TNF therapy (79–81) (CEBM 2b), and elevated CRP, anemia, and leukocytosis are predictors of relapse in CD in patients in whom withdrawal of medical therapy is being considered (82,83) (CEBM 2b). However, low CRP levels have been reported in patients with clinically-active (particularly ileal) disease (84) and, conversely, not all patients with elevated CRP have active disease (34). Further data are required before CRP can be recommended as a primary therapeutic target in clinical practice.

With regard to other biomarkers, there are few data to support the use of erythrocyte sedimentation rate as a means to monitor response to therapy, or as a therapeutic target in CD (85) (CEBM 5). Fecal calprotectin may be a useful adjunctive biomarker of response to therapy and to predict relapse, especially in Crohn's colitis (86) and after ileocectomy in CD (87). However, further data are required before its widespread use can be recommended (CEBM 3b (therapy), 2b (relapse)).

Thus, biomarkers may reflect residual intestinal inflammation, but endoscopy and/or magnetic resonance imaging should be performed before optimizing medical treatment. As such, biomarkers facilitate the monitoring of a patient rather than being a target for treatment *per se*. CRP and fecal calprotectin are adjunctive measures of inflammation for monitoring in CD, and failure of normalization (below lab-specific cutoff) using these tests should prompt further endoscopic or radiologic evaluation, irrespective of symptoms.

Patient-reported outcomes. A PRO is a report coming directly from a patient about how they function or feel in relation to a health condition and/or its therapy. While instruments such as the CDAI and Harvey Bradshaw Index incorporate patient-reported severity of symptoms and general well-being, they were not created according to the FDA guidance for development of a PRO, which involves generation of items from qualitative patient interviews and testing for reliability and responsiveness to changes in health status (36).

PROs have the potential to become a therapeutic target in IBD. To address the current lack of FDA-recognized PROs in CD, Khanna *et al.* (88) have developed a 2-item interim PRO composed of the abdominal pain and stool frequency domains of the CDAI. The group recommended that the primary PRO for CD should be resolution of abdominal pain and normalization of bowel habit. Given the variability in symptom expression in CD, the patient's individual treatment goals should also be addressed. The frequency of outcome assessment should be tailored to the patient's symptoms, with a minimum frequency of every 3 months until resolution and at least every 6–12 months after symptom resolution.

Various PROs are used in clinical trials and include quality of life (QoL), anxiety, depression, fatigue, work productivity, and disability (Table 3); however, outcome data with these PROs in IBD patients are limited and are not measured consistently. QoL instruments have been used frequently as secondary end points in clinical trials in IBD, including the combined use of a validated generic questionnaire and a validated disease-specific scale, which corre-

Table 3. Assessments for patient-reported outcomes in inflammatory bowel disease (Crohn's disease and ulcerative colitis)

Assessment	Instrument used
Quality of life	Clinical trials of biologics: combined use of a validated generic questionnaire (SF-36) and a validated disease-specific scale (IBD questionnaire (IBDQ) or short IBD questionnaire (SIBDQ)) (91,93,96–99) Other measures have included the Rating Form of Inflammatory Bowel Disease Patient Concerns (RFIPC) (90) and a second generic instrument, the European Quality of Life-5 dimensions (EQ-5D) (94)
Mood disorders	Tools available to assess depression and anxiety include the Zung Self-Rating Depression Scale, (111) the Center for Epidemiologic Studies Depression (CES-D) score, the Hospital Anxiety and Depression Scale-anxiety subscale (HADS-A) and -depression subscale (HADS-D), (110) and the Beck's Depression Inventory-II (BDI-II) (109)
Fatigue	The Checklist Individual Strength (120) and the Functional Assessment of Chronic Illness Therapy-Fatigue scale (FACIT-F) (118) are two screening tools that have been validated in studies of patients with IBD in correlating severity of fatigue with overall health-related quality of life as well as disease-specific quality of life The Multidimensional Fatigue Inventory (MFI) is another tool used in IBD to assess fatigue (116)
Work productivity	Measured by the Work Productivity and Activity Impairment Questionnaire (WPAI) (122)
IBD, inflammatory bowel disease.	

lates with disease activity for the assessment of biologics. The use of generic QoL instruments permits comparison with other conditions and health states, while disease-specific instruments focus on domains of greatest importance to patients with IBD (89–99) (CEBM 1b/2b).

Data evaluating mood disorders, including depression and anxiety, in IBD patients are scarce in clinical trials (100–102). Concomitant measurement of mood disorders is likely to be important given the subjective nature of PROs and the demonstrated effect of mood disorder on generic and disease-specific PROs (103–111) (CEBM 2b/3b).

Fatigue is the most common non-intestinal symptom reported by patients with IBD, and has a pervasive impact on subjective domains of health. Assessment tools for fatigue in IBD are listed in **Table 3**. Of note, the Functional Assessment of Chronic Illness Therapy-Fatigue scale score correlates with clinical disease activity in CD (112–121) (CEBM 2b/3b).

Work productivity can be measured by the Work Productivity and Activity Impairment Questionnaire (WPAI). The WPAI:CD, used in CD, is a validated questionnaire and correlates with disease activity (122). Although disability is generally recognized to occur in patients with IBD, it has never been measured. However, an IBD disability index is currently being validated (10,123–126) (CEBM 2b/3b).

Several scales have been developed over the past 20 years to assess PROs but none were developed according to the FDA guidance for PRO development, and thus cannot lead to PRO labeling (17,88,127). Once a consensus aimed at developing validated PROs based on the FDA guidance has been created, PROs are likely to be required by regulatory agencies as primary or major secondary end points in registration drug trials in patients with IBD.

Composite end point. Based on the recommendations above, a composite target of clinical/PRO remission (defined as resolution of abdominal pain and diarrhea/altered bowel habit) and endoscopic remission (resolution of ulceration at ileocolonoscopy or resolution of findings of inflammation on cross-sectional imaging in patients who cannot be adequately assessed with ileocolonoscopy), has been proposed (**Table 1**). In an analysis of a randomized, placebo-controlled trial of adalimumab in patients with early moderate-to-severe CD, 33% of adalimumab recipients achieved this end point after 3 months (128). *Post hoc* analysis of the Study of Biologic and Immunomodulator Naïve Patients In Crohn's Disease trial found that 23.1, 34.8, and 63% of biologic- and immunomodulator-naïve CD patients treated with azathioprine monotherapy, infliximab monotherapy, and azathioprine plus infliximab combination therapy, respectively, were able to achieve clinical remission and endoscopic remission after 6 months of treatment (129).

Ulcerative colitis

Clinical targets. Clinical targets in UC are the resolution of rectal bleeding and normalization of bowel habit. However, the resolution of symptoms alone is not a sufficient target, and objective evidence of inflammation of the bowel is necessary when making clinical decisions.

Data from the literature show that most clinical scoring systems for UC include stool frequency, bleeding, physician global assessment, and often abdominal pain. The Mayo score (and the partial version of the Mayo score without endoscopic subscore) (130) and the Rachmilewitz index (131) remain the clinical scores most commonly used during clinical trials (CEBM 1).

The Simple Clinical Colitis Activity Index (SCCAI) (132) and the Pediatric Ulcerative Colitis Activity Index (PUCAI) (133) clinical scoring systems have been generated using a prospective multistep approach and validated with a clear definition for clinical response and remission, but have been less used in trial setting. The PUCAI has been validated in adult and pediatric patients. For example, the PUCAI distinguishes three levels of clinical response: mild (10-point improvement), moderate (20-point improvement), and large (35-point improvement) (CEBM 1).

The SCCAI and the PUCAI broadly predict endoscopic disease activity; however, no clinical scores have been compared with the Ulcerative Colitis Endoscopic Index of Severity (UCEIS) (134) (CEBM 1).

Similar to CD, steroid-free clinical remission is achievable and sustainable in clinical trials (50,135–138). The PUCAI, the SCCAI, and the partial Mayo score can be used for clinical assessment. Clinical remission and clinical response are defined by a PUCAI of <10 points or a SCCAI of <1 or a partial Mayo score of

≤ 1 , and by a decrease of 20 for the PUCAI, at least 1 for the SCCAI or at least 2 for the partial Mayo score, respectively (133,139,140) (CEBM 1).

Endoscopic targets. The group recommends using the endoscopic Mayo score (rather than the UCEIS) at this time for UC, because although not validated, it is easy to use and has a well-established predictive value (141,142). The Mayo score has been found to have good intra- and inter-observer agreement among central readers (143). Based on the literature, a subscore of 0 or 1 should be considered as endoscopic remission (130) (CEBM 1a). Experts agreed that a Mayo endoscopic subscore of 0 is the optimal target, although there is insufficient information to recommend this target for all patients, and a subscore of 1 should be a minimum target (144,145). In the Active Ulcerative Colitis Trials-1 and -2 of infliximab in patients with moderate-to-severe UC, 25% of patients had achieved a Mayo endoscopy subscore of 0 by week 8, with 61% achieving a score of 0 or 1 (50). Using 54-week data from Active Ulcerative Colitis Trial-1, a Mayo endoscopy subscore of 0 was achieved by 33% of patients and a score of 0 or 1 was achieved by 50% of patients.

UCEIS is reliable for measuring endoscopic disease activity in UC and has been validated in an initial assessment (134,146) (CEBM 2b). Additional validation was carried out in an observer agreement study of seven expert central readers and demonstrated a strong correlation between UCEIS and global rating of endoscopic severity as well as excellent intra- and inter-reader agreement (147), however, additional validity testing is needed. Furthermore, although UCEIS 0/8 or 1/8 can be considered as remission, this has yet to be validated (146,148) (CEBM 2b). The UCEIS should be scored alongside endoscopic Mayo values in future trials, to further investigate the additional value and operating characteristics of the index (CEBM 5). Other scores, such as the Ulcerative Colitis Colonoscopic Index of severity, have also been developed (149,150).

Endoscopic assessment should be performed 3–6 months after the start of therapy in UC patients with symptoms.

Histologic targets. Attendees highlighted the importance of acknowledging that histopathology is a sensitive measure of inflammation in UC. Histological healing as a therapeutic end point is based on the premise that persistent inflammation in UC confers an increased risk of relapse, hospitalization, colectomy, and colorectal neoplasia, all of which lead to cumulative disability (49–55) (CEBM 3). Based on currently available evidence, several definitions of histological remission in UC have been proposed, although the most appropriate goal is yet to be determined (57–59) (CEBM 3). The group agreed that, at this time, histological healing is not a target in UC, because of limited evidence for its clinical utility in this patient group. As discussed earlier with CD, a paradigm shift in clinical thinking would be required to incorporate histology into routine management of UC patients, although it is more likely to occur with UC than with CD, owing to diffuse involvement of the mucosa, as well as the absence of neutrophils or basal plasmacytosis predict a lower risk of relapse and better

long-term outcome (as demonstrated by cohort studies) (151,152) (CEBM 5).

Imaging targets. Cross-sectional imaging techniques currently have a restricted role in UC (62,153,154) (CEBM 2). Thus, imaging is not a target in these patients.

Biomarkers. The group concluded that biomarkers, including CRP and fecal calprotectin, are not targets in UC. The literature review showed insufficient evidence to recommend treatment optimization using biomarkers alone, with only a few randomized control trials evaluating serum biomarkers, such as CRP, albumin, or hemoglobin, as end points in UC (155) (CEBM 1b). A small number of prospective randomized trials using clinical disease activity scores or colectomy rates as end points have shown that a reduction in CRP corresponds with treatment response (156–158) (CEBM 1b). In addition, CRP and albumin have been shown to be predictive of colectomy (a surrogate for treatment failure) in several studies (32,159,160), but the correlation of CRP levels with mucosal healing is modest (161–163) (CEBM 2b). Fecal calprotectin has been shown to correlate well with mucosal healing (16,164,165) (CEBM 2b). While fecal calprotectin has only recently been used as an end point in randomized controlled trials, it has been shown to correlate with response to induction therapy and to be predictive of loss of response to maintenance therapy (86,164–167) (CEBM 1b).

Nonetheless, biomarkers, such as CRP and fecal calprotectin, should only be used as adjunctive measures to facilitate the monitoring of a patient rather than being a target for treatment *per se*. Failure of CRP or fecal calprotectin normalization (below lab-specific cutoff) should prompt further endoscopic evaluation, irrespective of symptoms.

Patient-reported outcomes. As in CD, PROs are an important assessment outcome in UC and are likely to be required by regulatory agencies as primary or major secondary end points in registration drug trials in the future. The ultimate PRO in UC (as with CD above) is improvement in QoL (89). To achieve this, the primary PRO target for UC should be resolution of rectal bleeding and normalization of bowel habit; the patient's individual goals should also be addressed. The frequency of outcome assessment should be tailored to the patient's symptoms, with a minimum of every 3 months until resolution, and at least every 6–12 months after symptom resolution.

PRO assessment tools for IBD are presented in **Table 3**, and are similar between CD and UC. Data specific to UC shows that Functional Assessment of Chronic Illness Therapy-Fatigue scale score correlates with clinical disease activity, and with erythrocyte sedimentation rate, CRP, and hematocrit (118) (CEBM 2b/3b).

Composite end point. Based on the recommendations above, a composite target of clinical/PRO remission (defined as resolution of rectal bleeding and diarrhea/altered bowel habit) and endoscopic remission (defined as resolution of friability and ulceration at flexible sigmoidoscopy or colonoscopy) has been proposed (**Table 2**). Extrapolation of data from clinical

trials evaluating biological therapy (infliximab, adalimumab, or golimumab) in adults with moderate-to-severe active UC suggests that between 27 and 59% of patients who respond to induction therapy would meet this target after 1 year of treatment (50,168,169).

DISCUSSION

In recent decades, treatment goals in IBD have evolved greatly; we have moved from focusing on clinical response to the need to look beyond symptoms and achieve better outcomes associated with healing of mucosal lesions. In this context, the management of IBD is a challenge, with ambitious goals and expectations from both the physicians' and the patients' perspective, and there is a need to develop international recommendations that help these goals to be met in clinical practice.

The 12 recommendations presented here are based on a combination of published evidence, as identified through a systematic literature review, and expert opinion. Most statements received a very high level of agreement (average: 7.4–8.7 out of 10) and, since the attendees represented a number of different countries worldwide, this agreement suggests notable international recognition on the topics considered.

The group agreed that the ultimate goal in both CD and UC must be to restore the patient's QoL, because IBD can have a profound impact on PROs such as QoL, disability, or fatigue (17,170,171). The primary therapeutic target to achieve this in both UC and CD patients must be to achieve the composite end point of clinical/symptomatic and endoscopic remission, while also addressing the patient's individual goals (11,12). In the context of our recommendations, we define clinical remission as resolution of rectal bleeding and normalization of bowel habit in UC, and resolution of abdominal pain and normalization of bowel habit in CD. Endoscopic remission is defined as the resolution of friability and ulceration at flexible sigmoidoscopy or colonoscopy for UC, and resolution of ulceration at ileocolonoscopy in CD. Based on the data from randomized clinical trials, it is estimated that between one and two thirds of IBD patients should be able to meet these composite end points with intensive therapy, depending on their duration of disease and prior treatment history (50,128,129,168,169).

No randomized controlled trials have compared an intervention approach based on clinical symptoms only with an approach also targeting healing of the mucosa. However, considerable indirect evidence supports the notion that mucosal healing is associated with a reduced risk of relapse, fewer surgeries, fewer hospitalizations, and steroid tapering (49–55,172,173).

For UC, the Mayo endoscopic subscore has been the most commonly used tool in clinical trials for the evaluation of treatment efficacy in terms of endoscopic remission. This approach is therefore supported by the current evidence base, with mucosal healing defined as a subscore of 0–1 (130,137). On initial evaluation, the UCEIS appears to be reliable and should bring greater consistency to endoscopic assessment of disease severity in UC. The range of the score (0–8, rather than 0–3 in the Mayo endoscopic subscore)

may improve performance characteristics (148). However, full validation of remission and severity assessment, as well as the operating properties (responsiveness and reliability) of the UCEIS, is still in progress (146,174).

The group agreed that there is insufficient evidence to recommend treatment optimization by taking into account biomarkers alone. Biomarkers may reflect residual intestinal inflammation and can facilitate the monitoring of a patient, but endoscopy and/or imaging should be performed before optimizing medical treatment (175). Similarly, while histopathology is a sensitive way of assessing inflammation in UC, and is part of the overall assessment of the patient, it is not a treatment target. In addition, histological remission is not a target in clinical practice for CD. While cross-sectional imaging is an evolving technology, there is currently considerably more experience with the use of colonoscopy to evaluate mucosal inflammation. However, where evaluation of mucosal inflammation is not possible through colonoscopy (for example, proximal small bowel or perianal disease), imaging is a useful tool.

The recommendations provided as part of the STRIDE initiative are supported by a combination of published evidence, evaluated through a systematic review of the literature, and expert opinion, having involved a panel of international specialists in IBD. In contrast to RA, diabetes mellitus, and hypertension, there is not yet incontrovertible evidence in IBD that the implementation of these recommendations will lead to an altered disease course, so additional prospective data are required. In the meantime, there remains a need to take a case-by-case approach in clinical practice that considers disease severity, disease history, and prognostic factors. IBD is a heterogeneous disease, and different targets may need to be developed according to disease type. It is also critical to recognize that the “target” itself may not be achieved in certain patient types, and any recommendations should allow for this. For instance, similar to RA, there may be pre-existing structural damage before treatment, and mucosal healing may be more difficult to achieve in longstanding CD than in early disease (176). Furthermore, the safety of a treat-to-target approach needs to be established, particularly in patients with low risk of disease progression, where even minor risks of intensive treatment and monitoring can outweigh benefits, or in patients who may be more vulnerable to drug toxicity (e.g., older patients, patients with a long duration of disease or those with existing infections or immunosuppressive disorders). Challenges of applying a treat-to-target strategy in routine clinical practice also require consideration, such as convincing patients of the potential benefit or need to achieve a particular target and encouraging sufficient adherence to an intensive treatment and monitoring regimen. Finally, there is a significant information gap relating to the cost-effectiveness and cost-utility of treating-to-target: any escalation of treatment or monitoring will incur immediate expense (both to the health-care system and to the patient), but there is little evidence as to whether this will offset downstream costs such as hospitalization, resource use, disability-associated costs, and loss of productivity.

In conclusion, following an international consensus and a comprehensive literature review, recommendations on goals for

treating patients with IBD in clinical practice are made available. These include composite end points for both UC and CD that define the treatment target and the frequency of assessment to reach this target. Importantly, there are notable gaps in the published evidence base for many of the treatment goals, meaning that recommendations are largely based on expert opinion. We now need to test whether implementation of therapeutic algorithms pursuing these targets in clinical practice have the potential to change both disease course and patients' QoL. Ideally, a trial should be initiated to compare symptom-driven assessment with endoscopy-driven decisions.

ACKNOWLEDGMENTS

Writing assistance was provided by Helena Williams of Leading Edge Medical Education, part of the Lucid Group, UK, who received funding assistance from IOIBD. AbbVie and Ferring were not involved in the development of this manuscript and the authors maintained complete control over its content. We thank Severine Vermeire, Siew Ng, Marco Greco, Edouard Louis, and Marion O'Connor for their assistance.

CONFLICT OF INTEREST

Guarantor of the article: Jean-Frederic Colombel, MD.

Specific author contributions: Members of the Executive Steering Committee who lead the overall process of generating and agreeing the targets: L. Peyrin-Biroulet, W. Sandborn, B.E. Sands, J.F. Colombel, and S. Hanauer; attended the voting meeting: L. Peyrin-Biroulet, W. Sandborn, B.E. Sands, W. Reinisch, R.V. Bryant, I. Dotan, M. Dubinsky, B. Feagan, G. Fiorino, R. Gearry, S. Krishnareddy, P.L. Lakatos, E.V. Loftus, P. Marteau, T.B. Murdoch, I. Ordás, R.H. Riddell, J. Ruel, D.T. Rubin, M. Samaan, C.A. Siegel, J. Stoker, S. Schreiber, S. Travis, J. Panes, S. Hanauer, and J.F. Colombel; were workshop leaders for the 'Clinical targets' section: G. Van Assche and P.L. Lakatos; workshop leaders for endoscopic targets: G. D'Haens and P. Marteau; workshop leaders for histological targets: S. Travis and R.H. Riddell; workshop leaders for imaging targets, with S. Danese as a key contributor to this workgroup: J. Panes and W. Bemelman; workshop leaders for biomarkers: R. Panaccione and M. Samaan; workshop leaders for patient-reported outcomes: B.E. Sands and P. Munkholm; contributed to the systematic literature reviews: R.V. Bryant, G. Fiorino, S. Krishnareddy, T.B. Murdoch, I. Ordás, J. Ruel, M. Samaan, G. Bouguen, S. O'Donnell, B. Pariente, and S. Winer. All authors contributed to the critical review and revision of each draft of the manuscript and all authors approved the final version for submission.

Financial support: The STRIDE program was initiated by the IOIBD and received a multi-sponsored educational grant from AbbVie and Ferring; it is driven by an independent Steering Committee.

Potential competing interests: LPB: consulting fees from Merck, Abbott, Janssen, Genentech, Mitsubishi, Ferring, Norgine, Tillots, Vifor, Shire, Therakos, Pharmacosmos, Pilège, BMS, UCB-pharma, Hospira, Celltrion, Takeda, Boehringer-Ingelheim, Lilly, and Pfizer. Lecture fees from Merck, Abbott, Janssen, Ferring, Norgine, Tillots, Vifor, Therakos, and HAC-pharma. WS: consulting fees and research

support from Abbott Laboratories, Janssen Biotech, UCB Pharma, and Takeda, as well as consulting fees from Merck/Schering-Plough. BES: honoraria for service on an advisory board for AbbVie, Millennium Pharmaceuticals, Takeda, Shire, Pfizer, Luitpold Pharmaceuticals, Janssen Research and Development, Prometheus Laboratories, Forest Research Institute, Salix, and Vedanta Biosciences. WR: speaker, consultant, and/or advisory board member for Abbott Laboratories, Abbvie, Aesca, Amgen, AM Pharma, Aptalis, Astellas, Astra Zeneca, Avaxia, Bioclinica, Biogen IDEC, Bristol-Myers Squibb, Cellerix, Chemocentryx, Celgene, Centocor, Celltrion, Danone Austria, Elan, Falk Pharma GmbH, Ferring, Galapagos, Genentech, Grünenthal, Inova, Janssen, Johnson & Johnson, Kyowa Hakko Kirin Pharma, Lipid Therapeutics, MedImmune, Millenium, Mitsubishi Tanabe Pharma Corporation, MSD, Novartis, Ocera, Otsuka, PDL, Pharmacosmos, Pfizer, Procter & Gamble, Prometheus, Robarts Clinical Trial, Schering-Plough, Setpointmedical, Shire, Takeda, Therakos, Tigenix, UCB, Vifor, Yakult, Zyngenia, and 4SC. GD'H: Advisor for Abbvie, Ablynx, Actogenix, Amgen, AM Pharma, Astra Zeneca, Avaxia, Bristol-Myers Squibb, Boehringer-Ingelheim, Celgene, Celltrion, Cosmo, Covidien, Elan, Ferring, Falk Pharma, Centocor/Janssen Biologics, Engene, Galapagos, Gilead, GlaxoSmithKline, Hospira, Medimetrics, Millenium/Takeda, Mitsubishi Pharma, MSD, Mundipharma, NovoNordisk, Otsuka, Pfizer, Protein Design Laboratories, Prometheus Laboratories/Nestle, Receptos, Robarts Clinical Trials, Salix, Sandoz, Setpoint, Shire, Teva, Tigenix, Tillots, UCB, Versant and Vifor and received speaker fees from Abbvie, Ferring, Janssen Biologics, MSD, Mundipharma, Norgine, Shire, Takeda, Tillots, UCB, and Vifor. ID: Consultant for Janssen, AbbVie, Takeda, Genentech, and Pfizer; speaker's fees from Ferring, Falk Pharma, Janssen, AbbVie, and MSD. MCD: consultancy for Abbvie, Janssen, Takeda, UCB, Genentech, Pfizer, Shire, Salix, and Prometheus Labs, research support from Janssen and Prometheus Labs. BF: consultant for Abbott/Abbvie, Actogenix, Albireo Pharma, Amgen, Astra Zeneca, Avaxia Biologics Inc., Avir Pharma, Axcan, Baxter Healthcare Corp., Boehringer-Ingelheim, Bristol-Myers Squibb, Calypso Biotech, Celgene, Elan/Biogen, EnGene, Ferring Pharma, Roche/Genentech, GiCare Pharma, Gilead, Given Imaging Inc., GSK, Ironwood Pharma, Janssen Biotech (Centocor), JnJ/Janssen, Kyowa Kakko Kirin Co Ltd., Lexicon, Lilly, Merck, Millennium, Nektar, Novonordisk, Pfizer, Prometheus Therapeutics and Diagnostics, Protagonist, Receptos, Salix Pharma, Serono, Shire, Sigmoid Pharma, Synergy Pharma Inc., Takeda, Teva Pharma, TiGenix, Tillots, UCB Pharma, Vertex Pharma, Warner-Chilcott, Wyeth, Zealand, and Zyngenia; speaker's bureau for Abbott/Abbvie, JnJ/Janssen, Takeda, Warner-Chilcott, UCB Pharma; advisory board member for Abbott/Abbvie, Amgen, Astra Zeneca, Avaxia Biologics Inc., Bristol-Myers Squibb, Celgene, Centocor Inc., Elan/Biogen, Ferring, JnJ/Janssen, Merck, Novartis, Novonordisk, Pfizer, Prometheus Laboratories, Protagonist, Salix Pharma, Takeda, Teva, TiGenix, Tillots Pharma AG, UCB Pharma; grant/research support from Abbott/Abbvie, Amgen, Astra Zeneca, Bristol-Myers Squibb (BMS), Janssen Biotech (Centocor), JnJ/Janssen, Roche/Genentech, Millennium, Pfizer, Receptos, Santarus, Sanofi, Tillots, and UCB Pharma. GF: consultant and advisory board member for MSD, Takeda, AbbVie, and Janssen. RG:

speaker's fees, consultancies, ad boards, travel support from AbbVie, Janssen, Hospira, and Baxter. PLL: speaker for AbbVie, Eisai, Ferring, and MSD. In the past 2 years he has consulted for AbbVie, Celltrion, Eisai, Hospira, Ferring, MSD, and Takeda and has received unrestricted research grants from AbbVie and MSD. EVL: consultancy honoraria from AbbVie, Janssen, UCB, Takeda, MedImmune, and Immune Pharmaceuticals; research support from AbbVie, Janssen, UCB, Takeda, Genentech, Pfizer, Bristol-Myers Squibb, GlaxoSmithKline, Amgen, Santarus, and Robarts Clinical Trials. PM: honoraria for consultancy, and lecture and travel fees from AbbVie, Biocodex, Danone, Dr Falk, Ferring, Ipsen, Merck Médication Familiale, MSD, Nestlé, and Vifor Pharma. P Munkholm: Consultant and speaker for Tillotts, AbbVie, MSD, Takeda, Dr Falk, Shire, AstraZeneca, Swedish Orphan, Ferring, and Pharmacosmos. IO: speaker's fees from AbbVie and MSD. RP: consultant and/or lecture fees from AbbVie, Amgen, AstraZeneca, Axcen Pharma, Biogen Idec, Bristol-Myers Squibb, Centocor, ChemoCentryx, Eisai Medical Research, ELAN, Ferring, Genentech, GlaxoSmithKline, Janssen, MSD, Millennium, Ocera Therapeutics, Otsuka America Pharmaceuticals, Pfizer, Shire Pharmaceuticals, Prometheus, Schering Plough, Synta Pharmaceuticals, Teva, UCB Pharma, and Warner Chilcott. JR: consultant for Janssen, AbbVie, and Takeda. DTR: Consultant for Prometheus Pharmaceuticals, AbbVie, UCB, Janssen, and Takeda, grant support from Prometheus Pharmaceuticals and UCB. CAS: consultant/advisory board member for AbbVie, Lilly, Janssen, Salix, Pfizer, Prometheus, Takeda, and UCB; speaker for CME activities for AbbVie, Janssen, and Takeda. MSS: consulting fees, research support, and speaker fees received from Janssen, AbbVie, Takeda, and Prometheus. JS: consultant for Robarts Clinical Trials. SS: consultancy honoraria from AbbVie, Celltrion, Celgene, Ferring, Genentech, Hospira, Janssen, Robarts, UCB, and Takeda; speaker's fees from MSD and Falk Pharma. ST: former President of ECCO and conducted no industry consulting during his time as President, but since that time has acted as advisor to, lecturer for, or been in receipt of research support from AbbVie, AstraZeneca, Boehringer, Celgene, Cosmo, Ferring, Genentech, GSK, Novo Nordisk, NPS, Pfizer, Takeda, and Vifor. GvA: consultant for AbbVie, Takeda, Janssen, MSD, and Ferring; speaker's fees from AbbVie, MSD, Janssen, Warner-Chillcot, Ferring, Takeda, and Celltrion. SD: consultancy honoraria from AbbVie, AstraZeneca, Danone, MSD, Novo Nordisk, Takeda Millennium, Salix Pharmaceuticals, and Pfizer. JP: consultancy honoraria from AbbVie, Boehringer Ingelheim, Bristol-Myers Squibb, Galapagos, Janssen, MSD, Novo-Nordisk, Pfizer, Takeda, Topivert, and Tigenix. GB: speaker's fee from AbbVie, MSD, Ferring, and Norgine France. BP: lecture and travel fees from AbbVie and Ferring MSD. SH: consultant for AbbVie, Janssen, Shire, Salix, Takeda, UCB, Actavis, Ferring, and Prometheus, and on the speaker's bureau for AbbVie, Janssen, Salix, and Takeda. J-FC: consultant, advisory board member, or speaker for AbbVie, Bristol Meyers Squibb, Ferring, Genentech, Giuliani SPA, Given Imaging, Merck, Millenium Pharmaceuticals, Pfizer, Prometheus Laboratories, Sanofi, Schering Plough Corporation, Takeda, Teva Pharmaceuticals, and UCB Pharma.

Study Highlights

WHAT IS CURRENT KNOWLEDGE

- ✓ Recent studies focus on therapeutic targets beyond symptoms and underscore the need to treat mucosal lesions.
- ✓ In diseases such as rheumatoid arthritis, "treat-to-target" is used to improve outcomes and reduce the risk of end-organ damage.
- ✓ The optimal target for inflammatory bowel diseases has yet to be determined.

WHAT IS NEW HERE

- ✓ The Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE) group, under the auspices of the International Organization for the Study of Inflammatory Bowel Diseases (IOIBD), has developed 12 recommendations for treating to target in Crohn's disease (CD) and ulcerative colitis (UC).
- ✓ Clinical/patient-reported outcomes and endoscopic remission are recommended in clinical practice for both CD and UC.
- ✓ The implementation of these recommendations in clinical practice has the potential to change disease course and improve quality of life.

REFERENCES

1. Bernstein CN, Loftus EV Jr, Ng SC *et al.* Hospitalisations and surgery in Crohn's disease. *Gut* 2012;61:622–9.
2. Peyrin-Biroulet L, Loftus EV Jr, Colombel JF *et al.* The natural history of adult Crohn's disease in population-based cohorts. *Am J Gastroenterol* 2010;105:289–97.
3. Peyrin-Biroulet L, Loftus EV Jr, Colombel JF *et al.* Long-term complications, extraintestinal manifestations, and mortality in adult Crohn's disease in population-based cohorts. *Inflamm Bowel Dis* 2011;17:471–8.
4. Magro F, Rodrigues A, Vieira AI *et al.* Review of the disease course among adult ulcerative colitis population-based longitudinal cohorts. *Inflamm Bowel Dis* 2012;18:573–83.
5. Torres J, Billioud V, Sachar DB *et al.* Ulcerative colitis as a progressive disease: the forgotten evidence. *Inflamm Bowel Dis* 2012;18:1356–63.
6. Frolkis AD, Dykeman J, Negron ME *et al.* Risk of surgery for inflammatory bowel diseases has decreased over time: a systematic review and meta-analysis of population-based studies. *Gastroenterology* 2013;145:996–1006.
7. Jess T, Simonsen J, Jorgensen KT *et al.* Decreasing risk of colorectal cancer in patients with inflammatory bowel disease over 30 years. *Gastroenterology* 2012;143:375–81.
8. Vester-Andersen MK, Prosser MV, Jess T *et al.* Disease course and surgery rates in inflammatory bowel disease: a population-based, 7-year follow-up study in the era of immunomodulating therapy. *Am J Gastroenterol* 2014;109:705–14.
9. Pariente B, Cosnes J, Danese S *et al.* Development of the Crohn's disease digestive damage score, the Lemann score. *Inflamm Bowel Dis* 2011;17:1415–22.
10. Peyrin-Biroulet L, Cieza A, Sandborn WJ *et al.* Development of the first disability index for inflammatory bowel disease based on the international classification of functioning, disability and health. *Gut* 2012;61:241–7.
11. Peyrin-Biroulet L, Bressenot A, Kampman W. Histologic remission: the ultimate therapeutic goal in ulcerative colitis? *Clin Gastroenterol Hepatol* 2014;12:929–34.
12. Sandborn WJ, Hanauer S, Van Assche G *et al.* Treating beyond symptoms with a view to improving patient outcomes in inflammatory bowel diseases. *J Crohns Colitis* 2014;8:927–35.
13. Mosli MH, Feagan BG, Sandborn WJ *et al.* Histologic evaluation of ulcerative colitis: a systematic review of disease activity indices. *Inflamm Bowel Dis* 2014;20:564–75.

14. Jürgens M, Mahachie John JM, Cleynen I *et al*. Levels of C-reactive protein are associated with response to infliximab therapy in patients with Crohn's disease. *Clin Gastroenterol Hepatol* 2011;9:421–7.
15. Lewis JD. The utility of biomarkers in the diagnosis and therapy of inflammatory bowel disease. *Gastroenterology* 2011;140:1817–26.
16. D'Haens G, Ferrante M, Vermeire S *et al*. Fecal calprotectin is a surrogate marker for endoscopic lesions in inflammatory bowel disease. *Inflamm Bowel Dis* 2012;18:2218–24.
17. Williet N, Sandborn WJ, Peyrin-Biroulet L. Patient-reported outcomes as primary end points in clinical trials of inflammatory bowel disease. *Clin Gastroenterol Hepatol* 2014;12:1246–56.
18. Pincus T, Gibofsky A, Weinblatt ME. Urgent care and tight control of rheumatoid arthritis as in diabetes and hypertension: better treatments but a shortage of rheumatologists. *Arthritis Rheum* 2002;46:851–4.
19. Bouguen G, Levesque BG, Feagan BG *et al*. Treat to target: a proposed new paradigm for the management of Crohn's disease. *Clin Gastroenterol Hepatol* 2013, S1542–S3565.
20. Rachmani R, Slavachski I, Berla M *et al*. Treatment of high-risk patients with diabetes: motivation and teaching intervention: a randomized, prospective 8-year follow-up study. *J Am Soc Nephrol* 2005;16:S22–6.
21. Eeg-Olofsson K, Cederholm J, Nilsson PM *et al*. Glycemic and risk factor control in type 1 diabetes: results from 13,612 patients in a national diabetes register. *Diabetes Care* 2007;30:496–502.
22. Patel A, MacMahon S, Chalmers J *et al*. Effects of a fixed combination of perindopril and indapamide on macrovascular and microvascular outcomes in patients with type 2 diabetes mellitus (the ADVANCE trial): a randomised controlled trial. *Lancet* 2007;370:829–40.
23. Pearson TA. The epidemiologic basis for population-wide cholesterol reduction in the primary prevention of coronary artery disease. *Am J Cardiol* 2004;94:4f–8f.
24. Ridker PM, Danielson E, Fonseca FA *et al*. Rosuvastatin to prevent vascular events in men and women with elevated C-reactive protein. *N Engl J Med* 2008;359:2195–207.
25. Mora S, Musunuru K, Blumenthal RS. The clinical utility of high-sensitivity C-reactive protein in cardiovascular disease and the potential implication of JUPITER on current practice guidelines. *Clin Chem* 2009;55:219–28.
26. Pariente B, Cosnes J, Danese S *et al*. Development of the Crohn's disease digestive damage score, the Lémann score. *Inflamm Bowel Dis* 2011;17:1415–22.
27. Best WR, Becktel JM, Singleton JW *et al*. Development of a Crohn's disease activity index. National Cooperative Crohn's Disease Study. *Gastroenterology* 1976;70:439–44.
28. Thia KT, Sandborn WJ, Lewis JD *et al*. Defining the optimal response criteria for the Crohn's disease activity index for induction studies in patients with mildly to moderately active Crohn's disease. *Am J Gastroenterol* 2008;103:3123–31.
29. Harvey RF, Bradshaw JM. A simple index of Crohn's-disease activity. *Lancet* 1980;1:514.
30. Vermeire S, Schreiber S, Sandborn WJ *et al*. Correlation between the Crohn's disease activity and Harvey-Bradshaw indices in assessing Crohn's disease severity. *Clin Gastroenterol Hepatol* 2010;8:357–63.
31. Hyams JS, Ferry GD, Mandel FS *et al*. Development and validation of a pediatric Crohn's disease activity index. *J Pediatr Gastroenterol Nutr* 1991;12:439–47.
32. Turner D, Mack D, Leleiko N *et al*. Severe pediatric ulcerative colitis: a prospective multicenter study of outcomes and predictors of response. *Gastroenterology* 2010;138:2282–91.
33. Peyrin-Biroulet L, Reinisch W, Colombel JF *et al*. Clinical disease activity, C-reactive protein normalisation and mucosal healing in Crohn's disease in the SONIC trial. *Gut* 2014;63:88–95.
34. Jones J, Loftus EV Jr., Panaccione R *et al*. Relationships between disease activity and serum and fecal biomarkers in patients with Crohn's disease. *Clin Gastroenterol Hepatol* 2008;6:1218–24.
35. Modigliani R, Mary JY, Simon JF *et al*. Clinical, biological, and endoscopic picture of attacks of Crohn's disease. Evolution on prednisolone. Groupe d'Etude Thérapeutique des Affections Inflammatoires Digestives. *Gastroenterology* 1990;98:811–8.
36. Levesque BG, Sandborn WJ, Ruel J *et al*. Converging goals of treatment of inflammatory bowel disease from clinical trials and practice. *Gastroenterology* 2015;148:37–51.
37. Colombel JF, Sandborn WJ, Reinisch W *et al*. Infliximab, azathioprine, or combination therapy for Crohn's disease. *N Engl J Med* 2010;362:1383–95.
38. Hanauer SB, Feagan BG, Lichtenstein GR *et al*. Maintenance infliximab for Crohn's disease: the ACCENT I randomised trial. *Lancet* 2002;359:1541–9.
39. Panaccione R, Colombel JF, Sandborn WJ *et al*. Adalimumab maintains remission of Crohn's disease after up to 4 years of treatment: data from CHARM and ADHERE. *Aliment Pharmacol Ther* 2013;38:1236–47.
40. Daperno M, D'Haens G, Van Assche G *et al*. Development and validation of a new, simplified endoscopic activity score for Crohn's disease: the SES-CD. *Gastrointest Endosc* 2004;60:505–12.
41. Khanna R, Zou G, D'Haens G *et al*. Reliability among central readers in the evaluation of endoscopic findings from patients with Crohn's disease. *Gut*; advance online publication, 2 May 2015 [e-pub ahead of print].
42. Sipponen T, Savilahti E, Karkkainen P *et al*. Fecal calprotectin, lactoferrin, and endoscopic disease activity in monitoring anti-TNF-alpha therapy for Crohn's disease. *Inflamm Bowel Dis* 2008;14:1392–8.
43. Sipponen T, Savilahti E, Kolho KL *et al*. Crohn's disease activity assessed by fecal calprotectin and lactoferrin: correlation with Crohn's disease activity index and endoscopic findings. *Inflamm Bowel Dis* 2008;14:40–6.
44. Smith JP, Bingaman SI, Ruggiero F *et al*. Therapy with the opioid antagonist naltrexone promotes mucosal healing in active Crohn's disease: a randomized placebo-controlled trial. *Dig Dis Sci* 2011;56:2088–97.
45. Ferrante M, Colombel JF, Sandborn WJ *et al*. Validation of endoscopic activity scores in patients with Crohn's disease based on a post hoc analysis of data from SONIC. *Gastroenterology* 2013;145:978–86.
46. Ferrante M, de Hertogh G, Hlavaty T *et al*. The value of myenteric plexitis to predict early postoperative Crohn's disease recurrence. *Gastroenterology* 2006;130:1595–606.
47. Gecke K, Lowenberg M, Bossuyt P *et al*. Sal198 Agreement among experts in the endoscopic evaluation of postoperative recurrence in Crohn's disease using the Rutgeerts score. *Gastroenterology* 2014;146: S-227.
48. Reinisch W, Angelberger S, Petritsch W *et al*. Azathioprine versus mesalazine for prevention of postoperative clinical recurrence in patients with Crohn's disease with endoscopic recurrence: efficacy and safety results of a randomised, double-blind, double-dummy, multicentre trial. *Gut* 2010;59:752–9.
49. Baert F, Moortgat L, Van Assche G *et al*. Mucosal healing predicts sustained clinical remission in patients with early-stage Crohn's disease. *Gastroenterology* 2010;138:463–8.
50. Colombel JF, Rutgeerts P, Reinisch W *et al*. Early mucosal healing with infliximab is associated with improved long-term clinical outcomes in ulcerative colitis. *Gastroenterology* 2011;141:1194–201.
51. De Cruz P, Kamm MA, Prideaux L *et al*. Mucosal healing in Crohn's disease: a systematic review. *Inflamm Bowel Dis* 2013;19:429–44.
52. D'Haens G, Baert F, van Assche G *et al*. Early combined immunosuppression or conventional management in patients with newly diagnosed Crohn's disease: an open randomised trial. *Lancet* 2008;371:660–7.
53. Neurath MF, Travis SP. Mucosal healing in inflammatory bowel diseases: a systematic review. *Gut* 2012;61:1619–35.
54. Rutgeerts P, Diamond RH, Bala M *et al*. Scheduled maintenance treatment with infliximab is superior to episodic treatment for the healing of mucosal ulceration associated with Crohn's disease. *Gastrointest Endosc* 2006;63:433–42.
55. Schnitzler F, Fidler H, Ferrante M *et al*. Mucosal healing predicts long-term outcome of maintenance therapy with infliximab in Crohn's disease. *Inflamm Bowel Dis* 2009;15:1295–301.
56. Bryant RV, Winer S, Travis SP *et al*. Systematic review: histological remission in inflammatory bowel disease. Is 'complete' remission the new treatment paradigm? An IOIBD initiative. *J Crohns Colitis* 2014;8:1582–97.
57. Goldman H, Antonioli DA. Mucosal biopsy of the rectum, colon, and distal ileum. *Hum Pathol* 1982;13:981–1012.
58. Langner C, Magro F, Driessen A *et al*. The histopathological approach to inflammatory bowel disease: a practice guide. *Virchows Arch* 2014;464:511–27.
59. Magro F, Langner C, Driessen A *et al*. European consensus on the histopathology of inflammatory bowel disease. *J Crohns Colitis* 2013;7:827–51.
60. Geboes K, Riddell R, Ost A *et al*. A reproducible grading scale for histological assessment of inflammation in ulcerative colitis. *Gut* 2000;47:404–9.
61. Ordás I, Rimola J, Rodríguez S *et al*. Accuracy of magnetic resonance enterography in assessing response to therapy and mucosal healing in patients with Crohn's disease. *Gastroenterology* 2014;146:374–82.
62. Panés J, Bouhnik Y, Reinisch W *et al*. Imaging techniques for assessment of inflammatory bowel disease: joint ECCO and ESGAR evidence-based consensus guidelines. *J Crohns Colitis* 2013;7:556–85.

63. Panés J, Bouzas R, Chaparro M *et al.* Systematic review: the use of ultrasonography, computed tomography and magnetic resonance imaging for the diagnosis, assessment of activity and abdominal complications of Crohn's disease. *Aliment Pharmacol Ther* 2011;34:125–45.
64. Siddiki HA, Fidler JL, Fletcher JG *et al.* Prospective comparison of state-of-the-art MR enterography and CT enterography in small-bowel Crohn's disease. *AJR Am J Roentgenol* 2009;193:113–21.
65. Solem CA, Loftus EV Jr., Fletcher JG *et al.* Small-bowel imaging in Crohn's disease: a prospective, blinded, 4-way comparison trial. *Gastrointest Endosc* 2008;68:255–66.
66. Ordas I, Rimola J, Rodriguez S *et al.* Accuracy of magnetic resonance enterography in assessing response to therapy and mucosal healing in patients with Crohn's disease. *Gastroenterology* 2014;146:374–82.
67. Rimola J, Ordas I, Rodriguez S *et al.* Magnetic resonance imaging for evaluation of Crohn's disease: validation of parameters of severity and quantitative index of activity. *Inflamm Bowel Dis* 2011;17:1759–68.
68. Rimola J, Rodriguez S, Garcia-Bosch O *et al.* Magnetic resonance for assessment of disease activity and severity in ileocolonic Crohn's disease. *Gut* 2009;58:1113–20.
69. Bosch OG, Ordas I, Rodriguez S *et al.* 90 Comparison of the impact of MRI and colonoscopy on management of Crohn's disease. *Gastroenterology* 2012;142:S-21–2.
70. Bruining DH, Loftus EV Jr., Ehman EC *et al.* Computed tomography enterography detects intestinal wall changes and effects of treatment in patients with Crohn's disease. *Clin Gastroenterol Hepatol* 2011;9:679–83.
71. Bruining DH, Siddiki HA, Fletcher JG *et al.* Benefit of computed tomography enterography in Crohn's disease: effects on patient management and physician level of confidence. *Inflamm Bowel Dis* 2012;18:219–25.
72. Takenaka K, Ohtsuka K, Kitazume Y *et al.* Comparison of magnetic resonance and balloon enteroscopic examination of the small intestine in patients with Crohn's disease. *Gastroenterology* 2014;147:334–42.
73. Hanauer SB, Sandborn WJ, Rutgeerts P *et al.* Human anti-tumor necrosis factor monoclonal antibody (adalimumab) in Crohn's disease: the CLASSIC-I trial. *Gastroenterology* 2006;130:323–33.
74. Kiss LS, Papp M, Lovasz BD *et al.* High-sensitivity C-reactive protein for identification of disease phenotype, active disease, and clinical relapses in Crohn's disease: a marker for patient classification? *Inflamm Bowel Dis* 2012;18:1647–54.
75. Lamireau T, Cézard JP, Dabadie A *et al.* Efficacy and tolerance of infliximab in children and adolescents with Crohn's disease. *Inflamm Bowel Dis* 2004;10:745–50.
76. Rutgeerts P, D'Haens G, Targan S *et al.* Efficacy and safety of retreatment with anti-tumor necrosis factor antibody (infliximab) to maintain remission in Crohn's disease. *Gastroenterology* 1999;117:761–9.
77. Stack WA, Mann SD, Roy AJ *et al.* Randomised controlled trial of CDP571 antibody to tumour necrosis factor- α in Crohn's disease. *Lancet* 1997;349:521–4.
78. Targan SR, Hanauer SB, van Deventer SJ *et al.* A short-term study of chimeric monoclonal antibody cA2 to tumor necrosis factor α for Crohn's disease. Crohn's Disease cA2 Study Group. *N Engl J Med* 1997;337:1029–35.
79. Reinisch W, Wang Y, Oddens BJ *et al.* C-reactive protein, an indicator for maintained response or remission to infliximab in patients with Crohn's disease: a post-hoc analysis from ACCENT I. *Aliment Pharmacol Ther* 2012;35:568–76.
80. Sandborn WJ, Feagan BG, Stoinov S *et al.* Certolizumab pegol for the treatment of Crohn's disease. *N Engl J Med* 2007;357:228–38.
81. Schreiber S, Rutgeerts P, Fedorak RN *et al.* A randomized, placebo-controlled trial of certolizumab pegol (CDP870) for treatment of Crohn's disease. *Gastroenterology* 2005;129:807–18.
82. Louis E, Mary JY, Vernier-Massouille G *et al.* Maintenance of remission among patients with Crohn's disease on antimetabolite therapy after infliximab therapy is stopped. *Gastroenterology* 2012;142:63–70.
83. Treton X, Bouhnik Y, Mary JY *et al.* Azathioprine withdrawal in patients with Crohn's disease maintained on prolonged remission: a high risk of relapse. *Clin Gastroenterol Hepatol* 2009;7:80–5.
84. Florin TH, Paterson EW, Fowler EV *et al.* Clinically active Crohn's disease in the presence of a low C-reactive protein. *Scand J Gastroenterol* 2006;41:306–11.
85. Pepys MB, Hirschfield GM. C-reactive protein: a critical update. *J Clin Invest* 2003;111:1805–12.
86. Gisbert JP, Bermejo F, Perez-Calle JL *et al.* Fecal calprotectin and lactoferrin for the prediction of inflammatory bowel disease relapse. *Inflamm Bowel Dis* 2009;15:1190–8.
87. De Cruz P, Kamm MA, Hamilton AL *et al.* Crohn's disease management after intestinal resection: a randomised trial. *Lancet* 2015;385:1406–17.
88. Khanna R, Zou G, D'Haens G *et al.* A retrospective analysis: the development of patient reported outcome measures for the assessment of Crohn's disease activity. *Aliment Pharmacol Ther* 2015;41:77–86.
89. Burisch J, Weimers P, Pedersen N *et al.* Health-related quality of life improves during one year of medical and surgical treatment in a European population-based inception cohort of patients with inflammatory bowel disease—an ECCO-EpiCom study. *J Crohns Colitis* 2014;8:1030–42.
90. Drossman DA, Leserman J, Li ZM *et al.* The rating form of IBD patient concerns: a new measure of health status. *Psychosom Med* 1991;53:701–12.
91. Guyatt G, Mitchell A, Irvine EJ *et al.* A new measure of health status for clinical trials in inflammatory bowel disease. *Gastroenterology* 1989;96:804–10.
92. Irvine EJ, Feagan B, Rochon J *et al.* Quality of life: a valid and reliable measure of therapeutic efficacy in the treatment of inflammatory bowel disease. Canadian Crohn's Relapse Prevention Trial Study Group. *Gastroenterology* 1994;106:287–96.
93. Irvine EJ, Thompson AK. The Short Inflammatory Bowel Disease Questionnaire: a quality of life instrument for community physicians managing inflammatory bowel disease. CCRPT Investigators. Canadian Crohn's Relapse Prevention Trial. *Am J Gastroenterol* 1996;91:1571–8.
94. Jenkinson C, Gray A, Doll H *et al.* Evaluation of index and profile measures of health status in a randomized controlled trial. Comparison of the Medical Outcomes Study 36-Item Short Form Health Survey, EuroQol, and disease specific measures. *Med Care* 1997;35:1109–18.
95. Lönnfors S, Vermeire S, Greco M *et al.* IBD and health-related quality of life—Discovering the true impact. *J Crohns Colitis* 2014;8:1281–6.
96. Love JR, Irvine EJ, Fedorak RN. Quality of life in inflammatory bowel disease. *J Clin Gastroenterol* 1992;14:15–19.
97. McColl E, Han SW, Barton JR *et al.* A comparison of the discriminatory power of the Inflammatory Bowel Disease Questionnaire and the SF-36 in people with ulcerative colitis. *Qual Life Res* 2004;13:805–11.
98. van der Have M, van der Aalst KS, Kaptein AA *et al.* Determinants of health-related quality of life in Crohn's disease: a systematic review and meta-analysis. *J Crohns Colitis* 2014;8:93–106.
99. Ware JE Jr., Sherbourne CD. The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. *Med Care* 1992;30:473–83.
100. Graff LA, Walker JR, Bernstein CN. Depression and anxiety in inflammatory bowel disease: a review of comorbidity and management. *Inflamm Bowel Dis* 2009;15:1105–18.
101. Mikocka-Walus AA, Turnbull DA, Moulding NT *et al.* Controversies surrounding the comorbidity of depression and anxiety in inflammatory bowel disease patients: a literature review. *Inflamm Bowel Dis* 2007;13:225–34.
102. Nahon S, Lahmek P, Duranc C *et al.* Risk factors of anxiety and depression in inflammatory bowel disease. *Inflamm Bowel Dis* 2012;18:2086–91.
103. Beck AT, Ward CH, Mendelson M *et al.* An inventory for measuring depression. *Arch Gen Psychiatry* 1961;4:561–71.
104. Lichtenstein GR, Bala M, Han C *et al.* Infliximab improves quality of life in patients with Crohn's disease. *Inflamm Bowel Dis* 2002;8:237–43.
105. Loftus EV, Feagan BG, Colombel JF *et al.* Effects of adalimumab maintenance therapy on health-related quality of life of patients with Crohn's disease: patient-reported outcomes of the CHARM trial. *Am J Gastroenterol* 2008;103:3132–41.
106. Loftus EV Jr., Guerin A, Yu AP *et al.* Increased risks of developing anxiety and depression in young patients with Crohn's disease. *Am J Gastroenterol* 2011;106:1670–7.
107. Minderhoud IM, Samsom M, Oldenburg B. Crohn's disease, fatigue, and infliximab: is there a role for cytokines in the pathogenesis of fatigue? *World J Gastroenterol* 2007;13:2089–93.
108. Reinisch W, Sandborn WJ, Bala M *et al.* Response and remission are associated with improved quality of life, employment and disability status, hours worked, and productivity of patients with ulcerative colitis. *Inflamm Bowel Dis* 2007;13:1135–40.
109. Zhang CK, Hewett J, Hemming J *et al.* The influence of depression on quality of life in patients with inflammatory bowel disease. *Inflamm Bowel Dis* 2013;19:1732–9.
110. Zigmond AS, Snaith RP. The hospital anxiety and depression scale. *Acta Psychiatr Scand* 1983;67:361–70.
111. Zung WW. The Depression Status Inventory: an adjunct to the Self-Rating Depression Scale. *J Clin Psychol* 1972;28:539–43.

112. Beck A, Bager P, Jensen PE *et al*. How fatigue is experienced and handled by female outpatients with inflammatory bowel disease. *Gastroenterol Res Pract* 2013;2013:153818.
113. Czuber-Dochan W, Ream E, Norton C. Review article: Description and management of fatigue in inflammatory bowel disease. *Aliment Pharmacol Ther* 2013;37:505–16.
114. Graff LA, Clara I, Walker JR *et al*. Changes in fatigue over 2 years are associated with activity of inflammatory bowel disease and psychological factors. *Clin Gastroenterol Hepatol* 2013;11:1140–6.
115. Jelsness-Jorgensen LP, Bernklev T, Henriksen M *et al*. Chronic fatigue is associated with increased disease-related worries and concerns in inflammatory bowel disease. *World J Gastroenterol* 2012;18:445–52.
116. Romberg-Camps MJ, Bol Y, Dagnelie PC *et al*. Fatigue and health-related quality of life in inflammatory bowel disease: results from a population-based study in the Netherlands: the IBD-South Limburg cohort. *Inflamm Bowel Dis* 2010;16:2137–47.
117. Smets EM, Garssen B, Bonke B *et al*. The Multidimensional Fatigue Inventory (MFI) psychometric qualities of an instrument to assess fatigue. *J Psychosom Res* 1995;39:315–25.
118. Tinsley A, Macklin EA, Korzenik JR *et al*. Validation of the functional assessment of chronic illness therapy-fatigue (FACIT-F) in patients with inflammatory bowel disease. *Aliment Pharmacol Ther* 2011;34:1328–36.
119. van Langenberg DR, Gibson PR. Factors associated with physical and cognitive fatigue in patients with Crohn's disease: a cross-sectional and longitudinal study. *Inflamm Bowel Dis* 2014;20:115–25.
120. Vogelaar L, van't Spijker A, van Tilburg AJ *et al*. Determinants of fatigue in Crohn's disease patients. *Eur J Gastroenterol Hepatol* 2013;25:246–51.
121. Yellen SB, Cella DE, Webster K *et al*. Measuring fatigue and other anemia-related symptoms with the Functional Assessment of Cancer Therapy (FACT) measurement system. *J Pain Symptom Manage* 1997;13:63–74.
122. Reilly MC, Gerlier L, Brabant Y *et al*. Validity, reliability, and responsiveness of the work productivity and activity impairment questionnaire in Crohn's disease. *Clin Ther* 2008;30:393–404.
123. Høivik ML, Moum B, Solberg IC *et al*. Work disability in inflammatory bowel disease patients 10 years after disease onset: results from the IBSEN Study. *Gut* 2013;62:368–75.
124. Israeli E, Graff LA, Clara I *et al*. Low prevalence of disability among patients with inflammatory bowel diseases a decade after diagnosis. *Clin Gastroenterol Hepatol* 2014;12:1330–7.
125. Mandel MD, Bálint A, Lovász BD *et al*. Work disability and productivity loss in patients with inflammatory bowel diseases in Hungary in the era of biologics. *Eur J Health Econ* 2014;15:S121–8.
126. van der Valk ME, Mangen MJ, Leenders M *et al*. Risk factors of work disability in patients with inflammatory bowel disease—a Dutch nationwide web-based survey: work disability in inflammatory bowel disease. *J Crohns Colitis* 2014;8:590–7.
127. Spiegel BM, Hays RD, Bolus R *et al*. Development of the NIH Patient-Reported Outcomes Measurement Information System (PROMIS) gastrointestinal symptom scales. *Am J Gastroenterol* 2014;109:1804–14.
128. Colombel JF, Rutgeerts PJ, Sandborn WJ *et al*. Adalimumab induces deep remission in patients with Crohn's disease. *Clin Gastroenterol Hepatol* 2014;12:414–22.
129. Colombel JF, Reinisch W, Mantzaris GJ *et al*. Randomised clinical trial: deep remission in biologic and immunomodulator naive patients with Crohn's disease—a SONIC post hoc analysis. *Aliment Pharmacol Ther* 2015;41:734–46.
130. Schroeder KW, Tremaine WJ, Ilstrup DM. Coated oral 5-aminosalicylic acid therapy for mildly to moderately active ulcerative colitis. A randomized study. *N Engl J Med* 1987;317:1625–9.
131. Rachmilewitz D. Coated mesalazine (5-aminosalicylic acid) versus sulphasalazine in the treatment of active ulcerative colitis: a randomised trial. *BMJ* 1989;298:82–6.
132. Walmsley RS, Ayres RC, Pounder RE *et al*. A simple clinical colitis activity index. *Gut* 1998;43:29–32.
133. Turner D, Otley AR, Mack D *et al*. Development, validation, and evaluation of a pediatric ulcerative colitis activity index: a prospective multicenter study. *Gastroenterology* 2007;133:423–32.
134. Travis SP, Schnell D, Krzeski P *et al*. Developing an instrument to assess the endoscopic severity of ulcerative colitis: the Ulcerative Colitis Endoscopic Index of Severity (UCEIS). *Gut* 2012;61:535–42.
135. Feagan BG, Rutgeerts P, Sands BE *et al*. Vedolizumab as induction and maintenance therapy for ulcerative colitis. *N Engl J Med* 2013;369:699–710.
136. Panaccione R, Ghosh S, Middleton S *et al*. Combination therapy with infliximab and azathioprine is superior to monotherapy with either agent in ulcerative colitis. *Gastroenterology* 2014;146:392–400.
137. Rutgeerts P, Sandborn WJ, Feagan BG *et al*. Infliximab for induction and maintenance therapy for ulcerative colitis. *N Engl J Med* 2005;353:2462–76.
138. Sandborn WJ, van Assche G, Reinisch W *et al*. Adalimumab induces and maintains clinical remission in patients with moderate-to-severe ulcerative colitis. *Gastroenterology* 2012;142:257–65.
139. Turner D, Seow CH, Greenberg GR *et al*. A systematic prospective comparison of noninvasive disease activity indices in ulcerative colitis. *Clin Gastroenterol Hepatol* 2009;7:1081–8.
140. Lewis JD, Lichtenstein GR, Deren JJ *et al*. Rosiglitazone for active ulcerative colitis: a randomized placebo-controlled trial. *Gastroenterology* 2008;134:688–95.
141. Feagan BG, Reinisch W, Rutgeerts P *et al*. The effects of infliximab therapy on health-related quality of life in ulcerative colitis patients. *Am J Gastroenterol* 2007;102:794–802.
142. Sandborn WJ, Rutgeerts P, Feagan BG *et al*. Colectomy rate comparison after treatment of ulcerative colitis with placebo or infliximab. *Gastroenterology* 2009;137:1250–60.
143. Feagan BG, Sandborn WJ, D'Haens G *et al*. The role of centralized reading of endoscopy in a randomized controlled trial of mesalazine for ulcerative colitis. *Gastroenterology* 2013;145:149–57.
144. Inoue N, Takabayashi K, Takayama T *et al*. P215 Endoscopic remission (Mayo score 0 rather than score 1) predicts long-term clinical remission in ulcerative colitis. *J Crohns Colitis* 2013;7:S95.
145. Yokoyama K, Kobayashi K, Mukae M *et al*. Clinical study of the relation between mucosal healing and long-term outcomes in ulcerative colitis. *Gastroenterol Res Pract* 2013;2013:192794.
146. Travis SP, Schnell D, Krzeski P *et al*. Reliability and initial validation of the ulcerative colitis endoscopic index of severity. *Gastroenterology* 2013;145:987–95.
147. Pola S, Feagan BG, Fahmy M *et al*. Tu1106 Observer agreement and construct validity in central endoscopic assessment of disease activity in ulcerative colitis. *Gastroenterology* 2013;144:S-763.
148. Levesque BG, Loftus EV, Panaccione R *et al*. P150 responsiveness of endoscopic indices in the evaluation of ulcerative colitis. *J Crohns Colitis* 2014;8:S124.
149. Samuel S, Bruining DH, Loftus EV Jr. *et al*. Validation of the ulcerative colitis colonoscopic index of severity and its correlation with disease activity measures. *Clin Gastroenterol Hepatol* 2013;11:49–54.
150. Thia KT, Loftus EV Jr., Pardi DS *et al*. Measurement of disease activity in ulcerative colitis: interobserver agreement and predictors of severity. *Inflamm Bowel Dis* 2011;17:1257–64.
151. Bitton A, Peppercorn MA, Antonioli DA *et al*. Clinical, biological, and histologic parameters as predictors of relapse in ulcerative colitis. *Gastroenterology* 2001;120:13–20.
152. Riley SA, Mani V, Goodman MJ *et al*. Microscopic activity in ulcerative colitis: what does it mean? *Gut* 1991;32:174–8.
153. Ordás I, Rimola J, Garcia-Bosch O *et al*. Diagnostic accuracy of magnetic resonance colonography for the evaluation of disease activity and severity in ulcerative colitis: a prospective study. *Gut* 2013;62:1566–72.
154. Oussalah A, Laurent V, Bruot O *et al*. Diffusion-weighted magnetic resonance without bowel preparation for detecting colonic inflammation in inflammatory bowel disease. *Gut* 2010;59:1056–65.
155. Sands BE, Tremaine WJ, Sandborn WJ *et al*. Infliximab in the treatment of severe, steroid-refractory ulcerative colitis: a pilot study. *Inflamm Bowel Dis* 2001;7:83–8.
156. Aceituno M, Garcia-Planella E, Heredia C *et al*. Steroid-refractory ulcerative colitis: predictive factors of response to cyclosporine and validation in an independent cohort. *Inflamm Bowel Dis* 2008;14:347–52.
157. Cacheux W, Seksik P, Lemann M *et al*. Predictive factors of response to cyclosporine in steroid-refractory ulcerative colitis. *Am J Gastroenterol* 2008;103:637–42.
158. Gordon FH, Hamilton MI, Donoghue S *et al*. A pilot study of treatment of active ulcerative colitis with natalizumab, a humanized monoclonal antibody to alpha-4 integrin. *Aliment Pharmacol Ther* 2002;16:699–705.
159. Henriksen M, Jahnsen J, Lygren I *et al*. C-reactive protein: a predictive factor and marker of inflammation in inflammatory bowel disease. Results from a prospective population-based study. *Gut* 2008;57:1518–23.
160. Kumar S, Ghoshal UC, Aggarwal R *et al*. Severe ulcerative colitis: prospective study of parameters determining outcome. *J Gastroenterol Hepatol* 2004;19:1247–52.

161. Osada T, Ohkusa T, Okayasu I *et al.* Correlations among total colonoscopic findings, clinical symptoms, and laboratory markers in ulcerative colitis. *J Gastroenterol Hepatol* 2008;23:S262–7.
162. Solem CA, Loftus EV Jr, Tremaine WJ *et al.* Correlation of C-reactive protein with clinical, endoscopic, histologic, and radiographic activity in inflammatory bowel disease. *Inflamm Bowel Dis* 2005;11:707–12.
163. Yoon JY, Park SJ, Hong SP *et al.* Correlations of C-reactive protein levels and erythrocyte sedimentation rates with endoscopic activity indices in patients with ulcerative colitis. *Dig Dis Sci* 2014;59:829–37.
164. De Vos M, Louis EJ, Jahnsen J *et al.* Consecutive fecal calprotectin measurements to predict relapse in patients with ulcerative colitis receiving infliximab maintenance therapy. *Inflamm Bowel Dis* 2013;19:2111–7.
165. Molander P, af Björkstén CG, Mustonen H *et al.* Fecal calprotectin concentration predicts outcome in inflammatory bowel disease after induction therapy with TNF α blocking agents. *Inflamm Bowel Dis* 2012;18:2011–7.
166. Osterman MT, Aberra FN, Cross R *et al.* Mesalamine dose escalation reduces fecal calprotectin in patients with quiescent ulcerative colitis. *Clin Gastroenterol Hepatol* 2014;12:1887–93.
167. Yamamoto T, Shiraki M, Bamba T *et al.* Fecal calprotectin and lactoferrin as predictors of relapse in patients with quiescent ulcerative colitis during maintenance therapy. *Int J Colorectal Dis* 2014;29:485–91.
168. Sandborn WJ, Feagan BG, Marano C *et al.* Subcutaneous golimumab maintains clinical response in patients with moderate-to-severe ulcerative colitis. *Gastroenterology* 2014;146:96–109.
169. Sandborn WJ, Colombel JF, D'Haens G *et al.* One-year maintenance outcomes among patients with moderately-to-severely active ulcerative colitis who responded to induction therapy with adalimumab: subgroup analyses from ULTRA 2. *Aliment Pharmacol Ther* 2013;37:204–13.
170. Cohen BL, Zoega H, Shah SA *et al.* Fatigue is highly associated with poor health-related quality of life, disability and depression in newly-diagnosed patients with inflammatory bowel disease, independent of disease activity. *Aliment Pharmacol Ther* 2014;39:811–22.
171. Sherman M, Tsynman DN, Kim A *et al.* Sustained improvement in health-related quality of life measures in patients with inflammatory bowel disease receiving prolonged anti-tumor necrosis factor therapy. *J Dig Dis* 2014;15:174–9.
172. Ardizzone S, Maconi G, Russo A *et al.* Randomised controlled trial of azathioprine and 5-aminosalicylic acid for treatment of steroid dependent ulcerative colitis. *Gut* 2006;55:47–53.
173. Wright R, Truelove SR. Serial rectal biopsy in ulcerative colitis during the course of a controlled therapeutic trial of various diets. *Am J Dig Dis* 1966;11:847–57.
174. Annese V, Daperno M, Rutter MD *et al.* European evidence based consensus for endoscopy in inflammatory bowel disease. *J Crohns Colitis* 2013;7:982–1018.
175. Fengming Y, Jianbing W. Biomarkers of inflammatory bowel disease. *Dis Markers* 2014;2014:710915.
176. Peyrin-Biroulet L, Billioud V, D'Haens G *et al.* Development of the Paris definition of early Crohn's disease for disease-modification trials: results of an international expert opinion process. *Am J Gastroenterol* 2012;107:1770–6.